

Jinit Patel

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EDUCATION

The Ohio State University

M.S. Aerospace Engineering

B.S. Aerospace Engineering, Minor in Computer Science

- GPA: 3.885 (Dean's List All Semesters)

Columbus, OH

Expected May 2027

May 2026

EXPERIENCE

Honda Aircraft Company

Propulsion & Fuel Systems Engineering Intern

May 2026 – August 2026

Greensboro, NC

- Performed internal geometry sensitivity analyses on a 17-chamber, 32-port flammable fluid ventilation network in Amesim using CFD-derived pressure boundary conditions across the flight envelope; identified the primary flow restriction and an orifice change that increased critical-zone airflow by 40% (3.5 → 5.2 exchanges/min).
- Owned the design-build-test cycle for a modular airflow test rig with 10 differential pressure channels, using CATIA, CNC machining, TIG welding, and 3D-printed ducting to generate model validation data.
- Automated extraction and post-processing of 416 mass flow rate outputs per simulation batch using Python, Excel/VBA, and MATLAB scripts, cutting manual engineering time by 55 minutes/batch.
- Applied FFT analysis and a 1 Hz, 4th-order low-pass filter to reduce ambient noise across test rig pressure data.

Vertiv

Mechanical Engineering Intern

May 2025 – September 2025

Delaware, OH

- Rapidly prototyped & iterated a patent-pending actuation lever using carbon-fiber composite FDM printing, validating torsional structural performance through ANSYS FEA.
- Developed and documented 75+ electromechanical components in Creo, delivering detailed drawings with GD&T/MBD to drive consistency across design, manufacturing, and quality product lifecycle phases.

The Ohio State University Aerospace Research Center

Undergraduate Research Intern

May 2024 – December 2024

Columbus, OH

- Saved \$10,000 in production costs by designing a scalable, DFM-optimized SLA-printed nozzle in SolidWorks.
- Achieved 15° jet deflection at Mach 0.9 while investigating plasma-actuated thrust vectoring to expand jet control authority in next-generation VTOL UAV applications; leveraging PIV & Schlieren for flow field visualization.
- Ran 2D RANS CFD simulations in ANSYS Fluent, validating the flow field within ±5% of experimental results.
- AIAA published and presented [research findings](#) at the 33rd Annual Ohio Space Grant Consortium Symposium.

PROJECTS

End-to-End RC Fixed-Wing Aircraft Development

December 2025 – Present

- Designing, 3D-printing, and testing a custom RC airplane, applying aero analyses (XFLR5, Fluent), iterating with LW-PLA, and soldering propulsion & control electronics (ESC, flight controller, servos, sensors, battery).

GE Aerospace - Gas Turbine Engine Aerodynamic Loss Modeling Capstone

August 2025 – May 2026

- Modeled aerodynamic losses in gas turbines by building a CFD database to train an AI/ML model for pressure loss prediction and engine design optimization, with wind tunnel validation using pressure transducers.

INVOLVEMENT

Aerospace Honor Society - Sigma Gamma Tau | Treasurer

December 2024 – Present

- Oversee financial planning & operations for Sigma Gamma Tau, managing a \$6,000 annual budget.

BSLI | NASA Student Launch Initiative

August 2022 – May 2024

- Fabricated rocket fuselage and fins by performing wet layups, CNC routing, and laser cutting to construct a 12-ft carbon fiber fuselage, contributing to three successful launches.
- Achieved a 50-ft increase in peak altitude through aerodynamic geometry optimization in OpenRocket.

TECHNICAL SKILLS

Design & Simulation: Creo, SolidWorks, CATIA, ANSYS Fluent & Mechanical, STAR-CCM+, Amesim, XFLR5

Fabrication: Additive Manufacturing (SLA/FDM), CNC Routing & Milling, TIG Welding, Sheet Metal Forming,

Composite Layups, Pressure Valves & Regulators, Soldering & Wiring, Riveting, Deburring, and Precision Measurement

Programs: Python (NumPy, PyTorch), MATLAB/Simulink, C++, Teamcenter, Agile, Cura, Bambu Studio, MS Office

Standards & Methods: FAA 14 CFR Part 23, GD&T (ASME Y14.5), DFM/DFA, Design Reviews, RCA